



R/W Protocol Specification

Added batch function for DESFire Card

DUALi Inc.

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1. Introduction

This document defines the USB and serial communication protocol between DUALi's readers and a host computer (Some readers support USB only, but some other readers and modules support RS-232 with same protocol. So this document includes the RS-232 also).

This document defines batch function for DESFire card only.

Duali' readers support TDES and 3KTDES algorithm for DESFire card.

AES algorithm is not supported yet.

The protocols in this document are all for developers using DESFire Card.

All Duali's reader doesn't have this protocol by default.

So you should check that your device has this protocol before you develop your project.

This document is dedicated for all readers and modules. So, when you use some reader product of DUALi, those readers have a possibility to return code UNKNOWN COMMAND ERROR(23, 0x17), it means your reader or module don't support this command.

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2. Interface Specification

2.1 Communication Protocol Frame Format

2.1.1 Command frame format

(To show a Hexadecimal number, "0x" is appended to the first of the number)

Name	STX	LEN-H	LEN-L	CMD	Data[n]	LRC
Values	0x02	0xHH	0xHH	0xHH	Data[n]	0xHH
Length.	1-byte	1-byte	1-byte	1-byte	n-byte	1-byte

(STX and LRC are for RS-232. In case of USB, these elements should be removed)

- STX : 0x02
- Data length = command (It must be 1) + Length of data.
LEN-H : Higher byte of data length
LEN-L : Lower byte of data length
- CMD : Command byte
- Data[n] : Data bytes
- LRC : XORing from LEN-H to Data[n]

2.1.2 Response frame format

Name	STX	LEN-H	LEN-L	Resp	Data[n]	LRC
Values	0x02	0xHH	0xHH	0xHH	Data[n]	0xHH
Length.	1-byte	1-byte	1-byte	1-byte	n-byte	1-byte

(STX and LRC are for RS-232. In case of USB, these elements should be removed)

- STX : 0x02
- Data length = response (it must be 1) + Length of data.
LEN-H : Higher byte of data length
LEN-L : Lower byte of data length
- Resp : Response code from reader
- Data[n] : Data Bytes
- LRC : XORing from LEN-H to Data[n]

2.1.3 RS-232 Protocol (option)

- Speed : 115200bps (except DE-ABM4 : 57600bps, DE-930/DE-950 :9600bps)
- Data: 8 bit
- Parity : No
- Stop : 1 bit

3. Command Definition

3.1.1 RF Find Card (Command = 0x4C, _DE_FIND_CARD)

This command detects card in the RF field.

■ Command frame

LEN-H	LEN-L	CMD	Data[0]	Data[1]	Data[2]	Data[3]
0x0004		0x4C	BRate	CID	NAD	Option

- Data[0]

Data[0]	Max Card Baud Rate Use same speed for TX and RX.
0x00	0x00 = 106 Kbps
0x01	0x01 = 212 Kbps
0x02	0x02 = 424 Kbps
0x03	0x03 = 848 Kbps
0x04	VHBR, 1.6 ~ 6.8Mbps

- Data[1] : CID

- Data[2] : NAD

- Data[3] : Option ('A': only A-type, 'B': only B-type, 'K': NFC Barcode, other: All)

■ Response frame

LEN-H	LEN-L	Resp	Data[0]	Data[1]	Data[2...N-1]
N+1		OK	Type	CSPD	CD[...],UID[0]...UID[3]

- Data[0]

Data[0]	RF Type
'M', 0x4D, 77	Mifare
'A', 0x41, 65	A type
'B', 0x42, 66	B type

- Data[1]: Card communication speed

- Data[2]: Card Data and Card UID. It is different each card.

- Response: OK(0x00), Err(pls refer to the response code).

■ Response frame for NFC Barcode

LEN-H	LEN-L	Resp	Data[0...N-1]
N+1		OK	16bytes or 32bytes UID

3.1.2 DESFire Authentication (Command = 0x3A, _DE_DESFIRE_AUTH)

This command performs DESFire authentication.

■ Command frame

LEN-H	LEN-L	CMD	Data[0]	Data[1~16] or Data[1~24]
0x0011 or 0x0019		0x3A	KeyNo	Key

- Data[0] : Key number to perform authentication.

- Data[1~n] : Key to perform authentication.

If your card uses TDES, length of key is 16.

If your card uses 3KTDES, length of key is 24

■ Response frame

LEN-H	LEN-L	Resp	Data[0~15] or Data[0~23]
N+1		OK	Session key

- Data[0 ~ n] : Generated session key from authentication.

- Response: OK(0x00), Err(pls refer to the response code).

3.1.3 DESFire Transparent (Command = 0x64, _DE_DESFIRE_TRANSPARENT)

This command is used to send data with computed CRC from a host to the card.

Host doesn't need to compute 2-byte CRC. Reader extracts 2-byte CRC data from card response data.

This command makes frame for sending to DESFire card.

■ Command frame

LEN-H	LEN-L	CMD	Data[0]	Data[1]	Data[2~n]
0XXXXX		0x64	Flag	CMD len	Data

- Data[0] : Flag for ciphering data and adding CRC.

■ Bit Definitions

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0	0	0	0	Check Length	Add CRC32	Add CRC16	Cipher mode

- Bit3 : Set to 0, if you want that the device check data length automatically when you read data or write data to the DesFire card. (Split Data size : 36byte)

Set to 1, if you want to not checking length in the device.

- Bit2 : Set to 0, the device doesn't add crc32 and padding

Set to 1, the device adds crc32 and padding

- Bit1 : Set to 0, the device doesn't add crc16 and padding

Set to 1, the device adds crc16 and padding

- Bit0 : Set to 0, the device doesn't encipher data to transmit

Set to 1, the device encipher data to transmit

- Data[1] : Length of command part.
- Data[2~n] : Data to transmit (from command to data)

EX] protocol to write data (0x1122334455) with encryption. (key length is 24)

Int len = 2;

```
Sbuf[len++] = 0x64;
Sbuf[len++] = 0x05; //set bit2 and bit0
Sbuf[len++] = 0x07;
// start command part
Sbuf[len++] = 0x3D; //write command of DESFire card
Sbuf[len++] = 0x00; //number of file
Sbuf[len++] = 0x00; //offset high
Sbuf[len++] = 0x00; //offset low
Sbuf[len++] = 0x00; //length to write
Sbuf[len++] = 0x00; //length to write
Sbuf[len++] = 0x05; //length to write (write 5byte data)
Sbuf[len++] = 0x11; //data to write
Sbuf[len++] = 0x22; // data to write
Sbuf[len++] = 0x33; // data to write
Sbuf[len++] = 0x44; // data to write
Sbuf[len++] = 0x55; // data to write
Sbuf[0] = len/256;
Sbuf[1] = len%256;
```

■ Response frame

LEN-H	LEN-L	Resp code	Resp data
N+1		OK	

- Resp code: OK(0x00), Err(pls refer to the response code).
- Resp data : Response data of a DESFire card.

4. Example

4.1.1 Card Authentication

■ Condition

- AID : 000000
- keyno : 00,
- key : 00 (3keyDES)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000000 : Select Application
 - DESFire Transparent CMD : 64
 - Flag : 00
 - CMD len : 04
 - Select Application CMD : 5A
 - AID : 000000
3. 3A00 : Authenticate
 - DESFire Authentication CMD : 3A
 - Keyno : 00
 - Key : 00

4.1.2 Create Application

■ Condition

- AID : 000001
- keyno : 00,
- key : 00 (3keyDES)
- Key Sett.1 : 0F (default)
- Key Sett.2 : 41 (3K3DES, No 2 byte file identifier, 1 key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001: Select Application
3. 3A00 : Authenticate
4. 640006CA0000010F41 : Create Application
 - DESFire Transparent CMD : 64
 - Flag : 00
 - CMD len 06
 - CreateApplaication CMD : CA
 - AID : 000001

• Key Sett.1 : 0F

Bit 7...4	Change Key Access Hold the Access Rights for changing application keys(ChangeKey command) -0x0: Application master key authentication is necessary to change any key(default) -0x1 .. 0xD: Authentication with the specified key is necessary to change any key. -0xE: Authentication with the key to be changed(same KeyNo) is necessary to change a key. -0xF: All Keys (except application master key, see Bit0) within this application are frozen
Bit 3	Configuration Changeable Codes whether a change of the application master key settings is allowed -0: configuration not changeable anymore (frozen). -1: this configuration is changeable if authenticated with the application master key (default)
Bit 2	Free create / delete without master key Codes whether application master key authentication is needed before CreateFile/DeleteFile -1: CreateFile/DeleteFile is permitted only with application master key authentication. -0: CreateFile/DeleteFile is permitted also without application master key authentication (default)
Bit 1	Free directory list access without master key Codes whether application master key authentication is needed for file directory access -0: Successful application master key authentication is required for executing the GetFileIDs, GetFileSettings and GetKeySettings commands. -1: GetFileIDs, GetFileSettings and GetKeySettings commands succeed independently of a preceding application master key authentication (default)
Bit 0	Allow change master key Codes whether the application master key is changeable -0: Application master key is not changeable anymore (frozen) -1: Application master key is changeable (authentication with the current application master key necessary, default)

• Key Sett.2 : 41

Bit 6...7	Indicates the crypto method of the application - 00: specifies DES and 3DES operation for the whole application - 01 specifies 3K3DES operation for the whole application
-----------	---

	- 10: specifies AES operation for the whole application
Bit 5	Indicates use of 2 byte ISO/IEC 7816-4 File Identifiers for files within the Application - 0: No 2 byte File Identifiers for files within the application supported - 1: 2 byte File Identifiers for files within the application supported
Bit 4	RFU
Bit 0...3	Number of Keys that can be stored within the application for cryptographic Purposes. A maximum 14 keys can be stored within an application of DESFire. One can also create an application with no keys!

4.1.3 Delete Application

■ Condition

- AID : 000001
- Keyno : 00,
- key : 00 (3keyDES)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640004DA000001 : Delete Application
 - DESFire Transparent CMD : 64
 - Flag : 00
 - CMD len 04
 - DeleteApplaication CMD : DA
 - AID : 000001

4.1.4 Create Std Data File (Plain)

4.1.4.1 Create

■ Condition

- AID : 000001
- keyno : 00,
- key : 00 (3keyDES)
- FileNo : 00
- Com Set : 00 (Plain communication)
- Access Rights : EEEE (Free Access)
- File Size : 0A0000 (10)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640008CD0000EEEE0A0000 : Create Std data File

- DESFire Transparent CMD : 64
- Flag : 00
- CMD len 08
- CreateStdDataFile CMD : CD
- FileNo : 00
- Com. Setting : 00

00	Plain communication
01	Plain communication secured by DES/3DES MACing
03	Fully DES/3DES enciphered communication

- Access Right : EEEE

Byte 3	Change AccessRights Access Key No: 00~0D, 0E(Free Access)
Byte 2	Read&Write Access (GetValue, Debit, LimitedCredit, Credit for value files) Access Key No: 00~0D, 0E(Free Access)
Byte 1	Write Access (GetValue, Debit, LimitedCredit for value files) Access Key No: 00~0D, 0E(Free Access)
Byte 0	Read Access (GetValue, Debit, Debit for value files) Access Key No: 00~0D, 0E(Free Access)

- File Size : 0A0000

4.1.4.2 Read

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640008BD000000000A0000 : Read Data

- DESFire Transparent CMD : 64
- Flag : 00
- CMD len 08
- Read CMD : BD

- FileNo : 00
- Offset : 000000
- Length : 0A0000

4.1.4.3 Write

- Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 6400083D000000000A00002222222222222222222222222222 : Write Data
 - DESFire Transparent CMD : 64
 - Flag : 00
 - CMD len 08
 - Write CMD :3D
 - FileNo : 00
 - Offset : 000000
 - Length : 0A0000
 - Data : 22222222222222222222222222222222

4.1.4.4 Delete

- Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640002DF00 : Delete File
 - DESFire Transparent CMD : 64
 - Flag : 00
 - CMD len 02
 - Delete CMD : DF
 - FileNo : 00

4.1.5 Create Std Data File (Encrypt)

4.1.5.1 Create

- Condition

- AID : 000001
- keyno : 00,

- key : 00 (3keyDES)
- FileNo : 00
- Com Set : 03 (Fully DES/3DES enciphered communication)
- Access Rights : 0000 (Access key No 00)
- File Size : 0A0000 (10)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640008CD000300000A0000 : Create Std data File

- DESFire Transparent CMD : 64
- Flag : 00
- CMD len 08
- CreateStdDataFile CMD : CD
- FileNo : 00
- Com. Setting : 03

00	Plain communication
01	Plain communication secured by DES/3DES MACing
03	Fully DES/3DES enciphered communication

- Access Right : 0000

Byte 3	Change AccessRights Access Key No: 00~0D, 0E(Free Access)
Byte 2	Read&Wirte Access (GetValue, Debit, LimitedCredit, Credit for value flies) Access Key No: 00~0D, 0E(Free Access)
Byte 1	Write Access (GetValue, Debit, LimitedCredit for value files) Access Key No: 00~0D, 0E(Free Access)
Byte 0	Read Access (GetValue, Debit, Debit for value files) Access Key No: 00~0D, 0E(Free Access)

- File Size : 0A0000

4.1.5.2 Read (3Key)

■ Command

1. 4C00000100 : DetectCard

2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640508BD000000000A0000 : Read File
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 08
 - Read CMD : BD
 - FileNo : 00
 - Offset : 000000
 - Length : 0A0000

4.1.5.3 Write (3Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 6405083D000000000A000022222222222222222222222222222222 : Write Data
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 08
 - Write CMD : 3D
 - FileNo : 00
 - Offset : 000000
 - Length : 0A0000
 - Data : 22222222222222222222222222222222

4.1.5.4 Read (2Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640308BD000000000A0000 : Read File
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 08

- Read CMD : BD
- FileNo : 00
- Offset : 000000
- Length : 0A0000

4.1.5.5 Write (2Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 6403083D000000000A00002222222222222222222222222222 : Write Data
6403083d000000000a00002222222222222222222222222222
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 08
 - Write CMD :3D
 - FileNo : 00
 - Offset : 000000
 - Length : 0A0000
 - Data : 22222222222222222222222222222222

4.1.6 Create Value File

4.1.6.1 Create

■ Condition

- AID : 000001
- keyno : 00,
- key : 00 (3keyDES)
- FileNo : 01
- Com Set : 03 (Fully DES/3DES enciphered communication)
- Access Rights : 0000 (Access key No 00)
- File Size : 0A0000 (10)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640012CC0103000000000000FFFFF7F2710000001 : Create Value File

0012CC0100eeee00000000FFFFFF7F2710000001

- DESFire Transparent CMD : 64
- Flag : 00
- CMD len 12
- CreateValueFile CMD : CC
- FileNo : 01
- Com. Setting : 03

00	Plain communication
01	Plain communication secured by DES/3DES MACing
03	Fully DES/3DES enciphered communication

- Access Right : 0000

Byte 3	Change AccessRights Access Key No: 00~0D, 0E(Free Access)
Byte 2	Read&Write Access (GetValue, Debit, LimitedCredit, Credit for value files) Access Key No: 00~0D, 0E(Free Access)
Byte 1	Write Access (GetValue, Debit, LimitedCredit for value files) Access Key No: 00~0D, 0E(Free Access)
Byte 0	Read Access (GetValue, Debit, Debit for value files) Access Key No: 00~0D, 0E(Free Access)

- Lower Limit : 00000000
- Upper Limit : FFFFFFFF
- Value : 27100000
- Limited Credit Enabled : 01

Bit 0	Enable "LimitedCredit" feature. Here '0' means that "LimitedCredit" is disabled and '1' enables this feature.
Bit 1	Enable "Free GetValue" feature, which allows free read access to the value file. Here '0' means that "Free GetValue" is disabled and '1' enables this feature. If the access rights are set to disable any reading, this feature cannot be used.

4.1.6.2 Get Value (3Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate

4. 6405026C01 : Get Value

- DESFire Transparent CMD : 64
- Flag : 05
- CMD len 02
- GetValue CMD :6C
- FileNo : 01

4.1.6.3 Credit (3Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 6405020C0110000000 : Credit
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 02
 - Credit CMD :0C
 - FileNo : 01
 - Value : 10000000

4.1.6.4 Debit (3Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640502DC0110000000 : Debit
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 02
 - Debit CMD :DC
 - FileNo : 01
 - Value : 10000000

4.1.6.5 Get Value (2Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 6403026C01 : Get Value
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 02
 - GetValue CMD :6C
 - FileNo : 01

4.1.6.6 Credit (2Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
- 4.6403020C0110000000 : Credit
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 02
 - Credit CMD :0C
 - FileNo : 01
 - Value : 10000000

4.1.6.7 Debit (2Key)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
- 4.640302DC0110000000 : Debit
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 02
 - Debit CMD :DC
 - FileNo : 01
 - Value : 10000000

4.1.6.8 Transaction

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640501C7 : Transaction
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 01
 - Transaction CMD : C7

4.1.7 Change Key

4.1.7.1 3Key -> 3Key

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640502C400 : Change Key
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 02
 - Transaction CMD : C4
 - Key no : 00
 - Key : 00

4.1.7.2 2Key -> 2Key

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000001 : Select Application
3. 3A00 : Authenticate
4. 640302C400 : Change Key
 - DESFire Transparent CMD : 64
 - Flag : 03

- CMD len 02
- Transaction CMD : C4
- Key no : 00
- Key : 00000000000000000000000000000000

4.1.7.3 3Key -> 2Key (Application 0)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000000 : Select Application
3. 3A00 : Authenticate
4. 640502C400 : Change Key
 - DESFire Transparent CMD : 64
 - Flag : 05
 - CMD len 02
 - Transaction CMD : C4
 - Key no : 00
 - Key : 00000000000000000000000000000000

4.1.7.4 2Key -> 3Key (Application 0)

■ Command

1. 4C00000100 : DetectCard
2. 6400045A000000 : Select Application
3. 3A00 : Authenticate
4. 640302C44000 : Change Key
 - DESFire Transparent CMD : 64
 - Flag : 03
 - CMD len 02
 - Transaction CMD : C4
 - Key no : 40
 - Key : 00